

William Henry Melvin

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EDUCATION

James Madison University

Harrisonburg, VA

Bachelor of Science, Integrated Science & Technology (Concentration: Manufacturing) | Minor: Global Supply Chain Management

Junior | Expected Graduation: May 2028

Relevant Coursework: Thermodynamics, Applied Physics, Applied Statistics, Sustainable Manufacturing Systems, Programming, Biotechnology, Energy Issues, Calculus I & II

ENGINEERING EXPERIENCE

Engineering Intern — Ryzing Technology / Staunton, VA

May 2026 – August 2026

- Performed an ROI analysis on a fabric-cutting machine for Casualty Collection Point (CCP) / Casualty Protection Unit (CPU) production, modeling manual cutting time (859 cuts/set, ~41.7 min/set by hand) against the machine's rated 140 pieces/min throughput across a 100-set production volume — an estimated 85% reduction in cutting time; recommended against the purchase after projected payback fell short of the required return threshold.
- Created multi-view engineering drawings (flat, rolled, and isometric) with bills of materials for CCP/CPU circumferential components in Onshape; redesigned the circumferential pattern for repeatable, sewable manufacturing and fabricated a run of them by hand.
- Assisted in fabrication of air beam assemblies for CCP/CPU units, including measuring and adhesive bonding.
- Supported development and testing of the Navy's Inflatable Stability Air System (ISAS) — deployable air beams preventing a 1/8th-scale Amphibious Combat Vehicle (ACV) model from tipping during ship-to-shore transition — building a torsion pendulum rig to measure moment of inertia and radius of gyration at crew-loaded and gross-vehicle weight, and validating results against NSWC Carderock tolerance standards.
- Compiled ACV test findings into a presentation delivered to Navy personnel.
- Machined components to specification in support of NASA part requirements.
- Attended engineering program reviews with Navy, military, and NASA personnel as an observing intern.

ENGINEERING PROJECTS

Electronic Throttle Control System with Closed-Loop Speed Feedback

- Designed an Arduino-based electronic throttle control system, integrating Hall effect sensors and tachometers for closed-loop motor speed feedback.
- Redesigned the actuation mechanism from a rigid coupler to a pulley-and-cable system for smoother throttle response.
- Developed and tuned an embedded motion controller (TFT interface, SD-card-based CSV motion profiles) driving a servo-actuated mount, achieving fully repeatable operation in bench testing; further testing is on hold pending a replacement power supply after the shared unit failed while powering wave-generation equipment during a concurrent ISAS test.

TECHNICAL SKILLS

CAD: Onshape

Manufacturing: Manual CNC, Manual Lathe, 3D Printing, Engineering Drawings

Software: Microsoft Excel, Word, PowerPoint, Microsoft 365

CERTIFICATIONS

Google Project Management • Microsoft 365 Fundamentals • Microsoft Excel • Microsoft Word • Microsoft PowerPoint • CPR/AED for Professional Rescuers and First Aid • Mandatory Controlled Unclassified Information (DoD) • Onshape: Sketching, Part Studios, Assemblies, 2D Drawings, Direct Editing

ADDITIONAL EXPERIENCE

Lifeguard — West Side Swim Club

Summers 2023 – 2025

- Worked full-time as a lifeguard for three consecutive summers, including service as acting assistant manager during manager absences.
- Ensured patron safety through proactive monitoring and emergency response.
- Maintained CPR/AED certification and delivered excellent customer service.